Data Dictionary for the MIMIC-III Clinical Database

1. Introduction to the MIMIC-III Clinical Database

The Medical Information Mart for Intensive Care III (MIMIC-III) database represents a substantial, openly accessible, and de-identified collection of data pertaining to patients who received care in the critical care units of Beth Israel Deaconess Medical Center (BIDMC) between the years 2001 and 2012.[1] This extensive resource encompasses a wide array of information, including patient demographics, vital sign measurements recorded at the bedside (approximately one data point per hour), laboratory test results, details of medical and surgical procedures, administered medications, notes documented by nurses and physicians, imaging reports, and data concerning out-of-hospital mortality.[1] The MIMIC-III database is designed to support a diverse spectrum of analytical investigations, spanning areas such as epidemiology, the refinement of clinical decision-making rules, and the development of electronic healthcare tools.[1]

MIMIC-III is an updated version of its predecessor, MIMIC-II v2.6, and includes several new classes of data, such as approximately 20,000 additional ICU admissions, physician progress notes, medication administration records, more complete demographic information, and codes from the Current Procedural Terminology (CPT) and Diagnosis-Related Group (DRG) coding systems.[1] The progression from MIMIC-II to MIMIC-III illustrates an ongoing effort to enrich the depth and breadth of critical care data available for research purposes. The incorporation of physician notes and medication records within MIMIC-III expands the possibilities for studying clinical reasoning and treatment strategies.

1.2 Importance of a Data Dictionary

A data dictionary serves as a fundamental guide for comprehending the organization, content, and meaning of each individual component within the MIMIC-III database.[10] It is essential for the accurate interpretation of dimensions, which are typically categorical variables that classify data (e.g., patient gender, admission type), and measures, which are quantitative variables that represent measurable quantities (e.g., heart rate, length of stay). Without a clear understanding of each data element, researchers face the risk of misinterpreting the information, potentially leading to flawed analyses and erroneous conclusions. The data dictionary acts as a crucial intermediary, bridging the gap between the raw data and the extraction of meaningful insights relevant to specific research inquiries.

2. Accessing the Official MIMIC-III Documentation and Data Dictionary

2.1 Navigating the PhysioNet Website

PhysioNet [1] stands as the primary online repository for the MIMIC-III dataset and its associated resources. Researchers seeking to utilize this database should navigate to the MIMIC-III project page hosted on the PhysioNet platform.[1] It is also important to note that the MIMIC-III dataset is accessible through major cloud computing platforms, namely Amazon Web Services (AWS) and

Google Cloud Platform [2], which enhances accessibility and provides substantial computational resources for conducting large-scale data analyses. PhysioNet acts as a central point of access for the MIMIC-III community, offering not only the dataset itself but also crucial documentation and potentially tutorials or illustrative code examples. The availability of the database on cloud platforms further streamlines the research process by allowing researchers to work with the data in scalable computing environments.

2.2 Exploring the MIT Laboratory for Computational Physiology (LCP) Website

The Medical Information Mart for Intensive Care (MIMIC) database was developed and is maintained by the Laboratory for Computational Physiology (LCP) at the Massachusetts Institute of Technology (MIT).[2] The official MIMIC website, hosted by MIT-LCP [12], serves as another valuable resource for researchers. In particular, the documentation section of the MIT-LCP website [2] often contains more detailed explanations and information regarding the design principles and specific features of the dataset. This website can be considered a significant supplementary resource to PhysioNet, potentially offering deeper insights into the database's structure and content.

2.3 Identifying the Official Data Dictionary and Table Schemas

The primary source for the MIMIC-III data dictionary and table schemas is the official documentation [2] available on the MIT-LCP website.[17] This documentation may be presented in various formats, including downloadable files such as PDF or CSV documents, or as interactive web pages that provide descriptions for each table within the database. For instance, a project report from SUNY Oswego [8] provides an example of a comprehensive data dictionary tailored to their specific implementation of MIMIC-III. However, it is important to recognize that the official documentation hosted on the MIT website remains the most authoritative and up-to-date source of information regarding the MIMIC-III data dictionary. While third-party resources can offer helpful perspectives and interpretations, the official documentation should always be the primary reference point for the most accurate details about the dataset's structure and content.

3. Understanding the Relational Structure of MIMIC-III

3.1 Overview of the 26 Tables

The MIMIC-III database is structured as a relational database, comprising approximately 26 distinct tables.[3] These tables can be broadly categorized based on the type of information they contain, including tables for patient identification, records of hospital admissions, details of ICU stays, a comprehensive collection of clinical events (such as physiological measurements, laboratory test results, medication administration records, and clinical notes), billing-related information, and dictionary tables that provide definitions for various codes and identifiers used throughout the database.[3] It is worth noting that the demo subset of the MIMIC-III database also adheres to a similar schema, with the notable exception of the NOTEEVENTS table, which is excluded from the demo version.[4] The relational structure of the database is designed to allow researchers to link information pertaining to individual patients across different aspects of their hospital and ICU experiences. Understanding these interconnections between tables is fundamental for conducting thorough and meaningful analyses of the data. Researchers planning to use the demo version should be aware of the absence of clinical notes, as this might impact the scope of their intended analyses.

3.2 Common Identifiers and Linking Tables

A crucial aspect of the MIMIC-III database is the use of specific identifiers that facilitate the linking of information across different tables. These key identifiers include:

- SUBJECT_ID: This unique identifier is assigned to each individual patient within the database.[3] It allows researchers to track a single patient's data across multiple hospital admissions and ICU stays.
- HADM_ID: This identifier is unique to each hospital admission of a patient.[3] It links specific events and information to a particular instance of a patient's hospitalization.
- ICUSTAY_ID: This identifier is specific to each stay in an intensive care unit.[3] A single hospital admission can involve multiple ICU stays, each with its own unique ICUSTAY_ID.
- ITEMID: This identifier is used to represent specific charted items or concepts, such as a particular vital sign measurement (e.g., heart rate) or a specific laboratory test.[3] The ITEMID acts as a key to dictionary tables that provide the description of the item.

Tables within the MIMIC-III database that have column names ending with _ID often function as foreign keys, establishing relationships with other tables in the database.[3] These identifiers are the fundamental elements that allow researchers to integrate different types of patient data, such as demographics, diagnoses, procedures, medications, and physiological measurements. Understanding the level of granularity for each identifier (whether it refers to a patient, a hospital admission, an ICU stay, or a specific measurement) is essential for performing accurate data aggregation and analysis. Researchers will frequently utilize these identifiers to join tables and build comprehensive datasets tailored to their specific research questions.

4. Detailed Exploration of Key Data Tables and Their Dimensions

4.1 Patient Demographics (PATIENTS)

The PATIENTS table contains fundamental demographic information about each unique patient included in the MIMIC-III database. Key columns within this table include:

- SUBJECT_ID: A unique identifier assigned to each patient.[8]
- GENDER: The self-reported gender of the patient.[5]
- DOB: The patient's date of birth. It is important to note that dates in the MIMIC-III database have been shifted to protect patient anonymity.[5]
- DOD: The patient's date of death, if applicable. Similar to the date of birth, this date is also shifted.[4]
- DOD_HOSP: The date of death that occurred within the hospital setting.[8]
- DOD_SSN: The date of death as recorded by the Social Security Administration.[8]

- EXPIRE_FLAG: A binary flag indicating whether the patient died during their hospital stay.[5]

This table provides essential baseline characteristics of the patient population. The fact that dates have been shifted for de-identification is a critical consideration for any research involving temporal analyses, as the relative time differences between events are preserved, but the absolute dates are not the actual dates. The presence of multiple death date fields offers different perspectives on patient mortality, allowing researchers to define this outcome based on their specific criteria.

4.2 Hospital Admissions (ADMISSIONS)

The ADMISSIONS table provides detailed information about each hospital admission recorded in the MIMIC-III database. Key columns in this table include:

- HADM_ID: A unique identifier for each hospital admission.[3]
- SUBJECT_ID: Links this admission record to the corresponding patient in the PATIENTS table.[8]
- ADMITTIME: The timestamp indicating when the patient was admitted to the hospital.[5]
- DISCHTIME: The timestamp indicating when the patient was discharged from the hospital.[5]
- DEATHTIME: The timestamp indicating when the patient died during this specific hospital admission, if applicable.[5]
- ADMISSION_TYPE: Categorical variable describing the type of admission, such as emergency or elective.[8]
- ADMISSION_LOCATION: The location within the hospital where the patient was initially admitted.[5]
- DISCHARGE_LOCATION: The location to which the patient was discharged from the hospital.[5]
- INSURANCE: The type of insurance coverage the patient had during the admission.[5]
- LANGUAGE: The primary language spoken by the patient.[5]
- RELIGION: The patient's religious affiliation.[5]
- MARITAL_STATUS: The patient's marital status at the time of admission.[5]
- ETHNICITY: The patient's self-reported ethnicity.[5]
- DIAGNOSIS: A free-text field containing the admitting diagnosis provided by the admitting clinician. It is important to note that this diagnosis is not typically coded using a standardized ontology like ICD-9.[5]
- HOSPITAL_EXPIRE_FLAG: A binary flag indicating whether the patient died during this particular hospital admission.[8]
- HAS_CHARTEVENTS_DATA: A flag indicating whether there is at least one observation recorded in the CHARTEVENTS table associated with this admission.[8]

This table provides essential context for each hospital stay, including the timing of admission and discharge, demographic details, and the initial reason for hospitalization. The free-text nature of the DIAGNOSIS field highlights the importance of using the DIAGNOSES_ICD table for standardized diagnostic coding. The flags for in-hospital mortality and the availability of chart events data are useful for filtering and focusing analyses on specific subsets of admissions.

4.3 ICU Stays (ICUSTAYS)

The ICUSTAYS table specifically focuses on the periods when patients were admitted to an intensive care unit. Key columns in this table include:

- ICUSTAY_ID: A unique identifier for each ICU stay.[3]
- HADM_ID: Links this ICU stay to the corresponding hospital admission in the ADMISSIONS table.[3]
- SUBJECT_ID: Links this ICU stay to the patient in the PATIENTS table.[3]
- FIRST_CAREUNIT: The name of the first ICU unit the patient was admitted to during this stay.[8]
- LAST_CAREUNIT: The name of the last ICU unit the patient was in during this stay.[8]
- FIRST_WARDID: The ward identifier of the first care unit.[8]
- LAST_WARDID: The ward identifier of the last care unit.[8]
- INTIME: The timestamp indicating when the patient was admitted to the ICU.[5]
- OUTTIME: The timestamp indicating when the patient was discharged from the ICU.[5]
- LOS: The length of stay in the ICU, measured in days.[5]

This table provides specific details about each patient's time within the ICU. The LOS is a critical measure for many research questions related to the utilization of ICU resources and patient outcomes following critical illness. The information about the care units can be valuable for investigating differences in treatment and outcomes across various types of ICUs.

4.4 Diagnoses (DIAGNOSES_ICD, D_ICD_DIAGNOSES)

The MIMIC-III database uses the International Classification of Diseases, Ninth Revision, Clinical Modification (ICD-9-CM) coding system for recording diagnoses. This information is primarily stored in two tables:

- **DIAGNOSES_ICD:** This table contains the specific ICD-9 diagnosis codes assigned to each hospital admission.
  - ROW_ID: A unique identifier for each row in the table.[8]
  - SUBJECT_ID: Links to the PATIENTS table.[8]
  - HADM_ID: Links to the ADMISSIONS table.[8]
  - SEQ_NUM: A sequence number indicating the order or priority of the diagnosis for a given admission.[8] A value of 1 typically indicates the primary diagnosis.
  - ICD9_CODE: The ICD-9 diagnosis code itself, stored as a string to preserve any leading zeros.[3]
- **D_ICD_DIAGNOSES:** This table serves as a dictionary, providing descriptions for the ICD-9 diagnosis codes found in the DIAGNOSES_ICD table.
  - ROW_ID: A unique identifier for each row.[8]
  - ICD9_CODE: The ICD-9 diagnosis code, which is the primary key of this table.[8]
  - SHORT_TITLE: A short, abbreviated description of the ICD-9 code.[8]

    - LONG_TITLE: A more detailed, longer description of the ICD-9 code.[8]

Researchers will typically join the DIAGNOSES_ICD table with the D_ICD_DIAGNOSES table using the ICD9_CODE to obtain the clinical meaning of the diagnoses recorded for each hospital admission. The SEQ_NUM column in DIAGNOSES_ICD is important for analyses that differentiate between primary and secondary diagnoses.

4.5 Procedures (PROCEDURES_ICD, CPTEVENTS, D_ICD_PROCEDURES, D_CPT)

MIMIC-III records medical and surgical procedures using two different coding systems: ICD-9-CM and Current Procedural Terminology (CPT). The relevant tables are:

- **PROCEDURES_ICD:** This table contains the ICD-9 procedure codes performed during each hospital admission.
    - ROW_ID: A unique identifier for each row.[8]
    - SUBJECT_ID: Links to the PATIENTS table.[8]
    - HADM_ID: Links to the ADMISSIONS table.[8]
    - SEQ_NUM: A sequence number indicating the order of the procedure during the admission.[8]
    - ICD9_CODE: The ICD-9 procedure code, stored as a string.[3]
- **CPTEVENTS:** This table contains the CPT codes for procedures performed, along with associated cost center information.
    - ROW_ID: A unique identifier for each row.[8]
    - SUBJECT_ID: Links to the PATIENTS table.[8]
    - HADM_ID: Links to the ADMISSIONS table.[8]
    - COSTCENTER: The cost center associated with the procedure.[8]
    - CPT_CD: The Current Procedural Terminology code.[8]
- **D_ICD_PROCEDURES:** This dictionary table provides descriptions for the ICD-9 procedure codes.
    - ROW_ID: A unique identifier for each row.[8]
    - ICD9_CODE: The ICD-9 procedure code, which is the primary key.[8]
    - SHORT_TITLE: A short description of the ICD-9 procedure code.[8]
    - LONG_TITLE: A longer, more detailed description of the ICD-9 procedure code.[8]
- **D_CPT:** This dictionary table provides information about the CPT codes.
    - ROW_ID: A unique identifier for each row and the primary key.[8]
    - CATEGORY: The category of the CPT code.[8]
    - SECTIONRANGE: The range of codes within the high-level section.[8]
    - SECTIONHEADER: The header of the section.[8]
    - SUBSECTIONRANGE: The range of codes within the subsection.[8]
    - SUBSECTIONHEADER: The header of the subsection.[8]
    - CODESUFFIX: A text element of the CPT code.[8]
    - MINCODEINSUBSECTION: The minimum code within the subsection.[8]
    - MAXCODEINSUBSECTION: The maximum code within the subsection.[8]

Researchers should be aware that procedures are recorded using both ICD-9 and CPT systems in MIMIC-III. The choice of which table to use will depend on the specific research question and the type of procedure being investigated. The dictionary tables provide the necessary context for understanding the meaning of these procedural codes.

4.6 Medications (PRESCRIPTIONS, INPUTEVENTS_CV, INPUTEVENTS_MV)

Information about medications administered to patients in MIMIC-III is found in three primary tables:

- **PRESCRIPTIONS:** This table provides a record of medication orders.
  - ROW_ID: A unique identifier for each row.[5]
  - SUBJECT_ID: Links to the PATIENTS table.[5]
  - HADM_ID: Links to the ADMISSIONS table.[5]
  - ICUSTAY_ID: Links to the ICUSTAYS table, but may be null if the prescription was outside an ICU stay.[8]
  - DRUG_NAME: The name of the prescribed drug.[5]
  - DRUG_GENERIC: The generic name of the drug.[8]
  - NDC: The National Drug Code, a unique 11-digit identifier for drug products.[5]
  - DOSE_VAL: The numerical value of the dose.[8]
  - DOSE_UNIT: The unit of measurement for the dose.[8]
  - ROUTE: The route of administration of the drug.[8]
  - STARTDATE: The date when the prescription started.[8]
  - ENDDATE: The date when the prescription ended.[8]
- **INPUTEVENTS_CV and INPUTEVENTS_MV:** These tables contain detailed records of all fluids and medications administered to patients, differentiated by the information system used for data capture. INPUTEVENTS_CV contains data from the Philips CareVue system (used between 2001 and 2008), while INPUTEVENTS_MV contains data from the iMDSoft MetaVision system (used between 2008 and 2012).[3] These tables include columns such as ITEMID (which links to the D_ITEMS table to identify the specific medication or fluid), STARTTIME, ENDTIME, AMOUNT (the total amount administered), AMOUNTUOM (the unit of measurement for the amount), RATE (the infusion rate), RATEUOM (the unit of measurement for the rate), ORDERID (identifier for a specific order), and LINKORDERID (which links different instances of the same order, for example, when the infusion rate is changed).[5]

Researchers interested in a general overview of prescribed medications may find the PRESCRIPTIONS table sufficient. However, for detailed analyses of medication administration, including the timing, dosage, and rate of infusion, the INPUTEVENTS_CV and INPUTEVENTS_MV tables provide more granular information. The LINKORDERID is particularly useful for tracking changes in medication orders over time.

4.7 Laboratory Events (LABEVENTS, D_LABITEMS)

The results of laboratory tests performed on patients are stored in the LABEVENTS table. This table is linked to the D_LABITEMS dictionary table, which provides information about each specific lab test.

- **LABEVENTS:**
  - ROW_ID: A unique identifier for each row.[8]
  - SUBJECT_ID: Links to the PATIENTS table.[8]
  - HADM_ID: Links to the ADMISSIONS table.[8]
  - ITEMID: Links to the D_LABITEMS table, identifying the specific laboratory test.[8]
  - CHARTTIME: The timestamp indicating when the laboratory test was performed or when the result was recorded.[5]
  - VALUE: The raw, often textual, result of the laboratory test.[8]
  - VALUENUM: The numeric value of the laboratory test result, if applicable.[8]
  - VALUEUOM: The unit of measurement for the laboratory test result.[8]
  - FLAG: An indicator of whether the laboratory test result was considered abnormal.[8]
- **D_LABITEMS:**
  - ROW_ID: A unique identifier for each row.[8]
  - ITEMID: A unique identifier for each laboratory item and the primary key of this table.[8]
  - LABEL: The descriptive label or name of the laboratory test.[8]
  - FLUID: The type of fluid associated with the test, such as blood or urine.[8]
  - CATEGORY: The broader category of the laboratory test, such as hematology or chemistry.[8]
  - LOINC_CODE: The Logical Observation Identifiers Names and Codes (LOINC) code for the laboratory test.[8] LOINC codes provide a standardized way to identify laboratory tests.

The LABEVENTS table contains a wealth of information about patients' laboratory results over the course of their hospital and ICU stays. The ITEMID allows researchers to link each result to its description in the D_LABITEMS table. The VALUENUM field is particularly useful for quantitative analyses, while the VALUE field may contain more detailed or textual results. The VALUEUOM is crucial for interpreting the magnitude of the measurements, although some inconsistencies in units have been noted in the dataset.[17]

4.8 Chart Events (CHARTEVENTS, D_ITEMS)

The CHARTEVENTS table is one of the largest and most comprehensive tables in the MIMIC-III database. It contains detailed, time-stamped observations that were recorded on patient charts. This table draws data from both the CareVue and MetaVision information systems.[3]

- **CHARTEVENTS:**
  - Key columns include ROW_ID, SUBJECT_ID, HADM_ID, ICUSTAY_ID, ITEMID (which links to the D_ITEMS table), CHARTTIME (the timestamp of the observation), VALUE (the recorded value, often textual), VALUENUM (the numeric value, if applicable), VALUEUOM (the unit of measurement), STORETIME (the timestamp when the observation was stored in the database), CGID (identifier of the caregiver who recorded the observation), STOPPED (indicates if a continuous infusion was stopped), NEWBOTTLE (indicates if a new bag of

  solution was started), and ISERROR (flag indicating a potential error in the recorded data).[8]
- **D_ITEMS:** This table serves as a dictionary for the ITEMIDs found in the CHARTEVENTS table, as well as in the INPUTEVENTS tables. It provides descriptions for non-laboratory-related charted items.[8]
  - Key columns include ROW_ID, ITEMID (the primary key), LABEL (the descriptive name of the item), ABBREVIATION, DBSOURCE (indicating whether the item originated from CareVue or MetaVision), LINKSTO (specifies which table the ITEMID refers to, e.g., 'chartevents'), CATEGORY, UNITNAME, PARAM_TYPE (e.g., solution or ingredient), and CONCEPTID.[8]

Due to its vast size and the wide variety of clinical data it contains, the CHARTEVENTS table is a primary resource for researchers studying physiological measurements, vital signs, and other bedside observations. The DBSOURCE column is important for understanding the data's origin, as there might be differences in how data was recorded or structured in the two systems. The LINKSTO column in D_ITEMS confirms that a particular ITEMID corresponds to data in the CHARTEVENTS table. Researchers will typically filter this table based on specific ITEMIDs to extract the data relevant to their research questions. It is also important to note that text data sourced from drop-down menus in the MetaVision system has been incorporated into this table.[3]

4.9 Note Events (NOTEEVENTS)

The NOTEEVENTS table contains the unstructured clinical notes documented by healthcare providers during patient admissions. This includes a variety of note types such as nurse notes, physician progress notes, and discharge summaries.[1]

- Key columns in this table include ROW_ID, SUBJECT_ID, HADM_ID, CHARTDATE (the date the note was written), CHARTTIME (the time the note was written), STORETIME (the time the note was stored in the database), CATEGORY (e.g., 'Discharge summary', 'Nursing'), DESCRIPTION (a more specific description of the note type, e.g., 'Report' or 'Addendum' for discharge summaries), CGID (identifier of the caregiver who wrote the note), and TEXT (the full text content of the clinical note).[3]

The NOTEEVENTS table offers a rich source of information that can be analyzed using Natural Language Processing (NLP) techniques to extract insights into patient conditions, treatments, and clinical decision-making that may not be explicitly captured in the structured data fields. For example, addendums to discharge summaries can be identified because their DESCRIPTION field is changed from 'Report' to 'Addendum'.[3] It is important to remember that the demo version of the MIMIC-III database does not include the NOTEEVENTS table.[4]

4.10 Dictionary Tables (D_ITEMS, D_LABITEMS, D_ICD_DIAGNOSES, D_ICD_PROCEDURES, D_CPT)

These dictionary tables play a crucial role in providing definitions and metadata for the various codes and identifiers used throughout the MIMIC-III database.[3] They act as essential reference points for understanding the meaning of the codes used in the event tables. Detailed descriptions of the individual dictionary tables and their respective columns have been provided in sections 4.4, 4.5, 4.7, and 4.8. Researchers will frequently join these dictionary tables with the event tables (e.g., CHARTEVENTS with D_ITEMS, LABEVENTS with D_LABITEMS, DIAGNOSES_ICD with D_ICD_DIAGNOSES, PROCEDURES_ICD with D_ICD_PROCEDURES, and CPTEVENTS with D_CPT) to translate the coded values into human-readable descriptions, which is fundamental for interpreting and analyzing the clinical data.

5. Identifying and Defining Key Measures and Their Derivations

5.1 Length of Stay (LOS)

The length of a patient's hospital stay can be calculated by finding the difference between the ADMITTIME and DISCHTIME columns in the ADMISSIONS table.[8] This provides the total duration of the patient's hospitalization. For the length of stay specifically within the ICU, the ICUSTAYS table directly provides this information in the LOS column, measured in days.[5] Length of stay is a fundamental metric in healthcare, often used as a measure of resource utilization and as an outcome variable in studies of hospital care.

5.2 Mortality

Patient mortality can be determined using several fields within the MIMIC-III database. The DOD column in the PATIENTS table indicates the date of death for a patient.[5] Additionally, the DEATHTIME column in the ADMISSIONS table records the time of death during a specific hospital admission, and the HOSPITAL_EXPIRE_FLAG in the same table is a binary indicator of whether the patient died during that admission.[5] These different fields allow researchers to analyze mortality from various perspectives, such as overall mortality within the patient cohort or in-hospital mortality for specific admissions.

5.3 Vital Signs

A variety of vital signs are recorded in the CHARTEVENTS table. These include measurements such as heart rate, systolic and diastolic blood pressure, respiratory rate, oxygen saturation (SpO2), and body temperature.[1] Each of these vital signs is associated with specific ITEMIDs in the CHARTEVENTS table, which can be linked to their descriptions in the D_ITEMS table. These time-series data are essential for monitoring a patient's physiological status and can be used to derive trends, identify critical events, and calculate various physiological scores.

5.4 Laboratory Values

The LABEVENTS table contains the results of a wide range of laboratory tests, such as glucose levels, creatinine levels, hemoglobin levels, and many others.[1] Similar to vital signs, each specific laboratory test is identified by a unique ITEMID that links to the description of the test in the D_LABITEMS table. These laboratory values provide objective measures of a patient's biochemical and physiological state, which are critical for diagnosis, monitoring treatment response, and understanding disease progression.

5.5 Derived Measures

Beyond the directly recorded measures, the MIMIC-III dataset allows for the derivation of numerous other clinically relevant measures. For example, a patient's age at the time of admission or at any specific point in their record can be calculated by subtracting their date of birth (DOB in the PATIENTS table) from the relevant timestamp (e.g., ADMITTIME from the ADMISSIONS table or CHARTTIME from the CHARTEVENTS or LABEVENTS tables).[5] While not directly provided in the raw dataset, various clinical severity scores, such as the Sequential Organ Failure Assessment (SOFA) score or the Simplified Acute Physiology Score (SAPS), can be calculated using data from the CHARTEVENTS and LABEVENTS tables based on established clinical algorithms. The ability to derive such measures enhances the analytical potential of the MIMIC-III dataset, allowing researchers to explore complex relationships and outcomes.

6. Understanding Medical Terminologies and Coding Systems

6.1 ICD-9 Coding System

The International Classification of Diseases, Ninth Revision, Clinical Modification (ICD-9-CM) is a coding system that was widely used in the United States to classify diagnoses [3] and procedures [3] associated with hospital care.[31] The ICD-9-CM system consists of a tabular list of numerical disease codes, an alphabetical index to these entries, and a classification system for surgical, diagnostic, and therapeutic procedures, also with an alphabetical index and tabular list.[30] ICD-9 diagnosis codes typically range from three to five digits, sometimes with decimal sub-classifications to provide greater specificity.[16] ICD-9 procedure codes generally consist of three or four numeric digits.[35] In MIMIC-III, ICD-9 codes are used for recording diagnoses in the DIAGNOSES_ICD table [3] and for procedures in the PROCEDURES_ICD table.[3] The database also includes dictionary tables, D_ICD_DIAGNOSES and D_ICD_PROCEDURES, which provide short and long textual descriptions for these ICD-9 codes.[8] A thorough understanding of the ICD-9 coding system is essential for researchers to accurately interpret the diagnostic and procedural information within MIMIC-III. Resources from the Centers for Disease Control and Prevention (CDC) [30] and other authoritative sources can provide detailed information on specific ICD-9 codes and their clinical meanings. Although the healthcare industry has largely transitioned to the ICD-10 coding system in more recent years [32], familiarity with ICD-9 remains necessary for working with the MIMIC-III dataset, given its data collection period.

6.2 CPT Coding System

Current Procedural Terminology (CPT®) is another significant medical coding system used in MIMIC-III, primarily for reporting medical, surgical, and diagnostic procedures and services performed by physicians and other healthcare professionals.[1] The primary purpose of CPT codes is to provide a standardized language for documenting and reporting medical procedures, which streamlines the process of billing and reimbursement, and enhances the accuracy and efficiency of healthcare data reporting.[39] CPT codes are predominantly five-digit numeric codes, although Category II and Category III codes utilize an alphanumeric format.[40] The CPT coding system is organized into three main categories: Category I codes describe established procedures and services, Category II codes are supplemental tracking codes used for performance measurement, and Category III codes are temporary codes for new and emerging technologies, procedures, and services.[40] In MIMIC-III, CPT codes are found in the CPTEVENTS table [8], and the D_CPT dictionary table provides additional information about these codes.[8] Researchers working with the procedural data in MIMIC-III should understand the structure and purpose of the CPT coding system. Detailed information about CPT codes, including their descriptions and guidelines for their use, can be obtained from the American Medical Association (AMA), which maintains the CPT code set.[39]

6.3 LOINC Codes

Logical Observation Identifiers Names and Codes (LOINC) is a coding system used to uniquely identify laboratory tests and other clinical observations.[7] In the MIMIC-III database, LOINC codes are included in the D_LABITEMS table for laboratory tests.[8] These codes provide a standardized way to refer to specific laboratory tests, which facilitates the comparison of lab results across different healthcare systems and research studies, thereby enhancing data interoperability.

7. Conclusion

7.1 Summary of the MIMIC-III Dataset and the Importance of the Data Dictionary

The MIMIC-III Clinical Database stands as an invaluable resource for researchers in the field of critical care medicine and biomedical informatics. Its extensive collection of de-identified patient data offers a unique opportunity to investigate a wide range of clinical questions. The data dictionary is an indispensable tool for navigating and understanding the complexities of this database. By providing detailed descriptions of each table, column, and the coding systems employed, the data dictionary enables researchers to accurately interpret the information and extract meaningful insights.

7.2 Recommendations for Effectively Using the Dataset and Its Documentation

To effectively utilize the MIMIC-III dataset, researchers should begin by thoroughly examining the official documentation available on the MIT Laboratory for Computational Physiology (LCP) website. This documentation serves as the most authoritative source of information about the database structure and content. It is highly recommended to frequently consult the dictionary tables (D_ITEMS, D_LABITEMS, D_ICD_DIAGNOSES, D_ICD_PROCEDURES, D_CPT) to fully comprehend the meaning of the codes and identifiers used in the various event tables. Given the temporal nature of much of the data, researchers should carefully consider the timestamps associated with different events and be mindful of the date shifting applied for de-identification when conducting longitudinal analyses. Exploring the existing body of literature that has utilized the MIMIC-III dataset, as well as engaging with community resources such as the MIMIC Code

Repository on GitHub [12], can provide valuable insights and practical code examples for data analysis. Finally, it is crucial for all researchers to adhere to the data use agreement associated with MIMIC-III and to maintain ethical considerations when working with de-identified patient data.[1]

Table: Overview of Key MIMIC-III Tables

Table Name	**Brief Description**	**Key Identifiers**	**Primary Data Content**
PATIENTS	Contains demographic information for each unique patient.	SUBJECT_ID	Gender, date of birth, date of death (if applicable), expire flag.
ADMISSIONS	Details of each hospital admission.	HADM_ID, SUBJECT_ID	Admission and discharge times, admission type and location, discharge location, insurance, ethnicity, admitting diagnosis, in-hospital mortality flag.
ICUSTAYS	Information about each ICU stay.	ICUSTAY_ID, HADM_ID, SUBJECT_ID	ICU admission and discharge times, length of stay in ICU, first and last care units.
DIAGNOSES_ICD	ICD-9 diagnosis codes for each hospital admission.	ROW_ID, SUBJECT_ID, HADM_ID, SEQ_NUM, ICD9_CODE	ICD-9 diagnosis codes and their sequence number.
PROCEDURES_ICD	ICD-9 procedure codes performed during each hospital admission.	ROW_ID, SUBJECT_ID, HADM_ID, SEQ_NUM, ICD9_CODE	ICD-9 procedure codes and their sequence number.
CPTEVENTS	CPT procedure codes performed during hospital admissions.	ROW_ID, SUBJECT_ID, HADM_ID, CPT_CD	CPT procedure codes and associated cost center.

PRESCRIPTIONS	Medication orders.	ROW_ID, SUBJECT_ID, HADM_ID, ICUSTAY_ID, DRUG_NAME	Drug name, generic name, NDC code, dosage, route, start and end dates.
INPUTEVENTS_CV/MV	Detailed records of fluids and medications administered (CareVue and MetaVision).	ROW_ID, SUBJECT_ID, HADM_ID, ICUSTAY_ID, ITEMID	Start and end times, amount, rate of administration for specific medications and fluids.
LABEVENTS	Results of laboratory tests.	ROW_ID, SUBJECT_ID, HADM_ID, ITEMID, CHARTTIME	Laboratory test results (value, numeric value, unit of measurement), abnormality flag.
CHARTEVENTS	Time-stamped observations recorded on patient charts.	ROW_ID, SUBJECT_ID, HADM_ID, ICUSTAY_ID, ITEMID, CHARTTIME	Vital signs, physiological measurements, and other charted data.
NOTEEVENTS	Unstructured clinical notes.	ROW_ID, SUBJECT_ID, HADM_ID, CHARTTIME	Text content of various types of clinical notes (e.g., discharge summaries, nurse notes).
D_ICD_DIAGNOSES	Dictionary for ICD-9 diagnosis codes.	ROW_ID, ICD9_CODE	Short and long descriptions of ICD-9 diagnosis codes.
D_ICD_PROCEDURES	Dictionary for ICD-9 procedure codes.	ROW_ID, ICD9_CODE	Short and long descriptions of ICD-9 procedure codes.
D_CPT	Dictionary for CPT codes.	ROW_ID	Category, section and subsection ranges and headers

			for CPT codes.
D_ITEMS	Dictionary for non-laboratory-related charted items.	ROW_ID, ITEMID	Labels, abbreviations, source database, linked table, category, unit name for charted items.
D_LABITEMS	Dictionary for laboratory-related items.	ROW_ID, ITEMID	Labels, fluid type, category, LOINC code for laboratory tests.

Table: Examples of Key Measures and Their Corresponding Tables

Measure Name	**Table(s) of Origin**	**Key Columns Involved in Derivation (if applicable)**
Hospital Length of Stay	ADMISSIONS	DISCHTIME - ADMITTIME
ICU Length of Stay	ICUSTAYS	LOS
In-Hospital Mortality	ADMISSIONS	HOSPITAL_EXPIRE_FLAG, DEATHTIME
Overall Mortality	PATIENTS	DOD
Heart Rate	CHARTEVENTS	ITEMID (specific to heart rate), VALUENUM, CHARTTIME
Systolic Blood Pressure	CHARTEVENTS	ITEMID (specific to systolic BP), VALUENUM, CHARTTIME
Diastolic Blood Pressure	CHARTEVENTS	ITEMID (specific to diastolic BP), VALUENUM, CHARTTIME
Oxygen Saturation (SpO2)	CHARTEVENTS	ITEMID (specific to SpO2),

		VALUENUM, CHARTTIME
Temperature	CHARTEVENTS	ITEMID (specific to temperature), VALUENUM, CHARTTIME
Glucose Level	LABEVENTS	ITEMID (specific to glucose), VALUENUM, CHARTTIME
Creatinine Level	LABEVENTS	ITEMID (specific to creatinine), VALUENUM, CHARTTIME
Age at Admission	PATIENTS, ADMISSIONS	ADMITTIME - DOB

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